

Informational Retrieval: Assignment 1

Embeddings System

1. Introduction

In this assignment, I built an embedding system built upon three classifier algorithms: (1) logistic regression, (2) random forest, and (3) linear SVM. These algorithms work in tandem with the sentence embedding model Sentence Transformers (all-mpnet-base-v2). The resulting system takes the user description of a historical weapon they see somewhere, but do not know the name of, as input. The system presents the most likely weapon based on the description.

2. Tools

The models were developed with the help of at least three artificial intelligence tools, namely Codex, ChatGPT, and Gemini. Codex was used as a preliminary coding tool, where I prompted the general idea of the system and continued refining it until I hit my token limit. Then, the system was further developed with the help of ChatGPT and, finally, Gemini. Meanwhile, this report was made without the help of any generative artificial intelligence tool.

3. Data

The data for this system were compiled from Wikipedia using a proprietary code for fetching data from the Wikipedia API. At first, the code generated only 1.000 lines from the weapons listed on the page "List of premodern combat weapons." Later, the code was expanded to be able to generate over 10.000 lines. However, after Gemini suggested that the data were too noisy, changes were made so the code fetched only 500-line worth of weapon descriptions.

4. Development

The system was developed with the help of various generative intelligence tools. The first tool used to build the system was Codex. Codex is an AI-powered coding tool that allows users to input instruction prompts. The tool will then generate codes accordingly. The free version of Codex limits users in terms of the amount of code it can provide. I encountered this very problem before I could finish the system.

The development then continued with ChatGPT. At this stage, the skeleton of the system was ready. The GPT-powered me just needed to flesh it out. However, the endeavor proved to be challenging. Both Codex and ChatGPT failed to capture the extent I wanted my system to cover. I wanted to build a system that can take user descriptions and provide them with a weapon's name in return. Codex did not seem

to want me to fetch too many articles from Wikipedia. Meanwhile, ChatGPT seemed to be averse to fetching the texts and kept leading me to fetch article titles instead.

After manually checking the generated custom data and codes, I turned to Gemini. This tool was more collaborative in my case, provided that I insisted on what I wanted after the said checking. That said, Gemini also seemed to be adamant in limiting the amount of data I should fetch from Wikipedia.

5. Reflection

Working with artificial intelligence tools was enjoyable, if not fun. I would actually suggest that new Language Technology students start their first semester with a similar project. As a former linguistics student, I have been struggling to keep up. I can imagine myself feeling more motivated if the project was tasked to me early in the academic year. "Hey, I will be able to create these kinds of tools soon (without AI)!"